MENU del GIORNO

1. GRAFICI di FUNZIONE E COMPOSIZIONE di FUNZIONI
2. INVERSA DI UNA FUNZIONE
3. NUMERI COMPLESSI

1) GRAFICI E COMPOSIZIONE

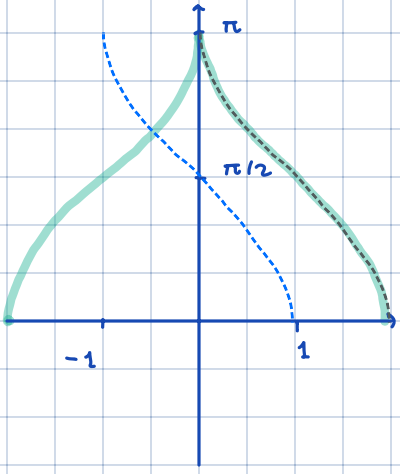
1.1)

$f(x) = \arccos x$

$f \circ g(x) = f(g(x))$

$\overset{!}{=} \arccos(g(x))$

$\overset{!}{=} \arccos(|x|-1)$



$h_0(x) = \arccos x$

$h_1(x) = h_0(x-1)$

$\overset{!}{=} \arccos(x-1)$

$h_2(x) = h_1(|x|)$

$\overset{!}{=} \arccos(|x|-1)$

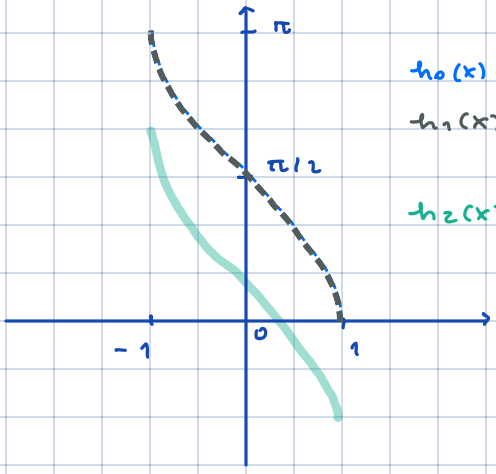
$y = \arccos(|x|-1)$

$g(x) = |x|-1$

$g \circ f(x) = g(f(x))$

$\overset{!}{=} |f(x)|-1$

$\overset{!}{=} |\arccos x|-1$



$h_0(x) = \arccos x$

$h_1(x) = |h_0(x)|$

$\overset{!}{=} |\arccos x|$

$h_2(x) = h_1(x)-1$

$\overset{!}{=} |\arccos x|-1$

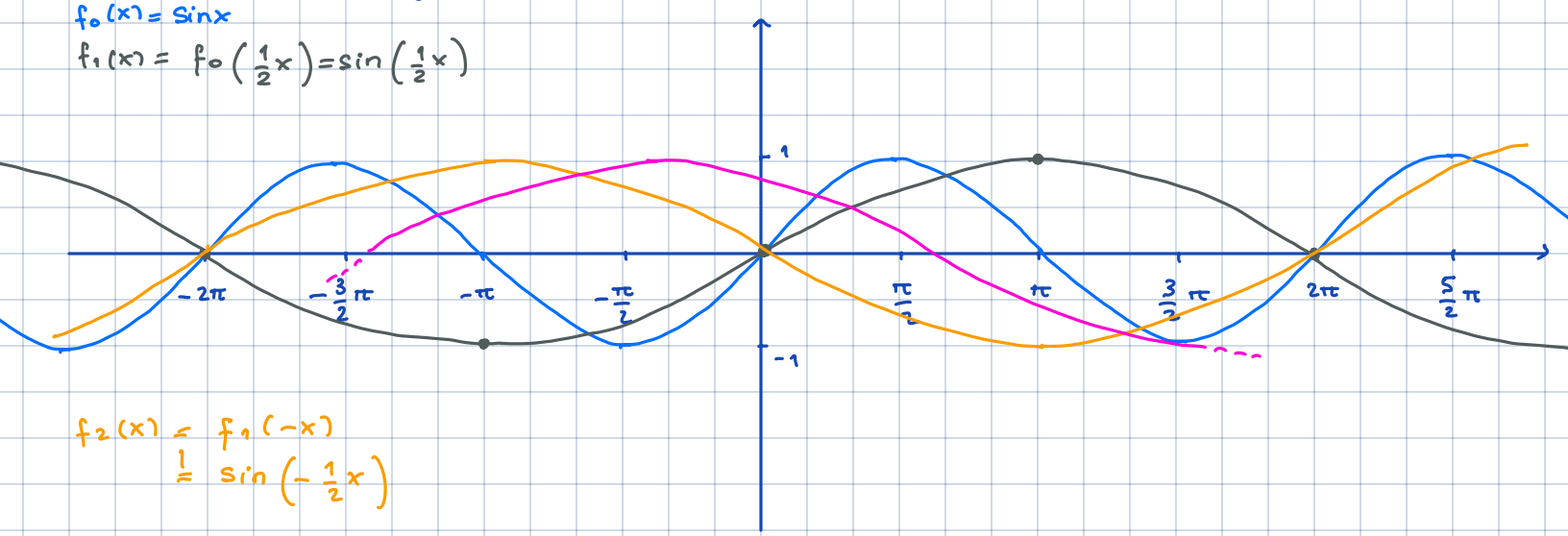
1.2

$f(x) = 2\sin\left(1-\frac{1}{2}x\right)+1$

$f_0(x) = \sin x$

$f_1(x) = f_0\left(\frac{1}{2}x\right) = \sin\left(\frac{1}{2}x\right)$

$f_1(\pi) = f_0\left(\frac{\pi}{2}\right)$



$f_2(x) = f_1(-x)$

$\overset{!}{=} \sin\left(-\frac{1}{2}x\right)$

$f_3(x) = f_2(x-2) = \sin\left(-\frac{1}{2}(x-2)\right) = \sin\left(1-\frac{1}{2}x\right)$

1.3)

$y = 2\sqrt{|x|-1}+1$

1.4) $f(x) = \sqrt{x+1}$, $g(x) = x^2 - 1$ (PROVETTA)

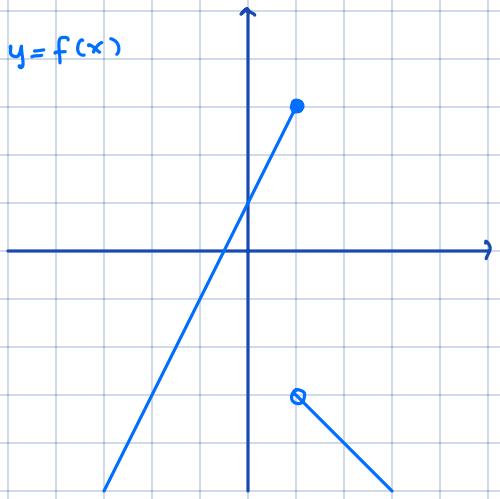
$$f \circ g(x) = f(g(x)) = \sqrt{g(x)+1} = \sqrt{x^2-1+1} = \sqrt{x^2} = |x|$$

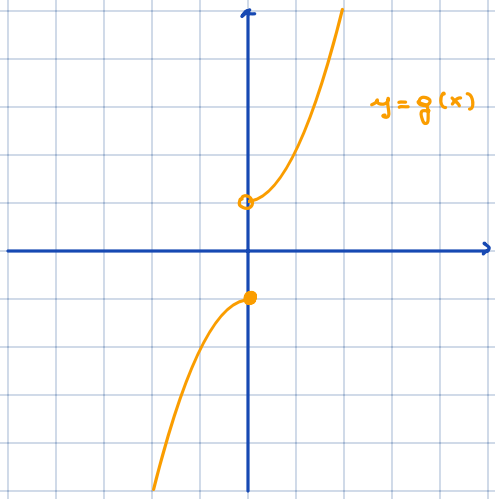
1.5) $f(x) = x - x^2$ $\quad g(y) = y^2 - y$

$$g \circ f(x) = g(f(x)) = f^2(x) - f(x) = (x-x^2)^2 - (x-x^2)$$
$$= x^2 + x^4 - 2x^3 - x + x^2$$
$$= x^4 - 2x^3 + 2x^2 - x$$

1.6) $$f(x) = \begin{cases} 2x+1 & , \ x \leq 1 \\ -x-2 & , \ x > 1 \end{cases} \qquad g(x) = \begin{cases} x^2+1 & , \ x > 0 \\ -x^2-1 & , \ x \leq 0 \end{cases}$$

$f \circ g(x) = ?$ $\qquad f(g(x)) = \begin{cases} 2g(x)+1 & , \ g(x) \leq 1 \\ -g(x)-2 & , \ g(x) > 1 \end{cases}$





OSS:
$g(x) > 1 \leftrightarrow x > 0$
$g(x) \leq 1 \leftrightarrow x \leq 0$

$$f \circ g(x) = f(g(x)) = \begin{cases} 2g(x)+1 & , \ g(x) \leq 1 \\ -g(x)-2 & , \ g(x) > 1 \end{cases} = \begin{cases} 2g(x)+1 & , \ x \leq 0 \\ -g(x)-2 & , \ x > 0 \end{cases}$$

$$g(x) = \begin{cases} x^2+1 & , \ x > 0 \\ -x^2-1 & , \ x \leq 0 \end{cases} \longrightarrow = \begin{cases} 2(-x^2-1)+1 & , \ x \leq 0 \\ -(x^2+1)-2 & , \ x > 0 \end{cases}$$

$$f(g(x)) = \begin{cases} 2(-x^2-1)+1 & , \ x \leq 0 \\ -(x^2+1)-2 & , \ x > 0 \end{cases}$$

2) FUNZIONE INVERSA

2.1 Calcolare l'inversa (e disegnare il grafico) di

$f: [1/2, +\infty) \to \mathbb{R}$ definita da $f(x) = x^2 - x + 1$

$y = x^2 - x + 1$

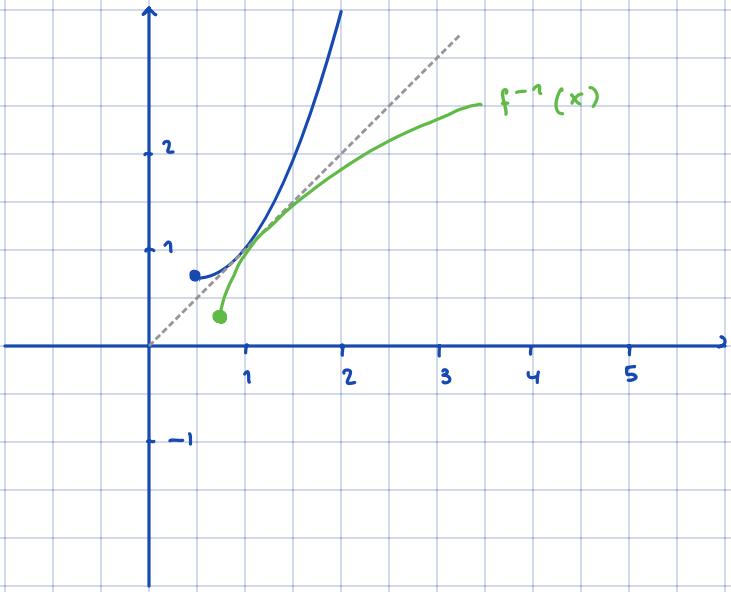
$x^2 - x + 1 - y = 0$

$$x_{1,2} = \frac{1 \pm \sqrt{1 - 4(1-y)}}{2} = \frac{1 \pm \sqrt{4y-3}}{2}$$

$$\to x = \frac{1 + \sqrt{4y-3}}{2} \quad \text{ACC. } x \geq \frac{1}{2}$$

$$\to f^{-1}(x) = \frac{1 + \sqrt{4x-3}}{2}$$

$$\to x = \frac{1 - \sqrt{4y-3}}{2} \quad \text{NON ACC } \left(x < \frac{1}{2}\right)$$



$f(x) = f^{-1}(x)$

$x^2 - x + 1 = \frac{1 + \sqrt{4x-3}}{2}$

$\Downarrow$

$\begin{cases} y = f(x) \\ y = x \end{cases}$

$x^2 - x + 1 = x \to x^2 - 2x + 1 = 0$

$(x-1)^2 = 0$

$x = 1$

2.2 $f(x) = e^{2x} + e^x$

$y = e^{2x} + e^x$ oss.: $y > 0$

$e^{2x} + e^x - y = 0$

$t = e^x$

$t^2 + t - y = 0$

$$t_{1,2} = \frac{-1 \pm \sqrt{1+4y}}{2}$$

$$\to t = \frac{-1 + \sqrt{1+4y}}{2} \to e^x = \frac{-1 + \sqrt{1+4y}}{2}$$

$$\to x = \log\left(\frac{-1 + \sqrt{1+4y}}{2}\right)$$

$$\to f^{-1}(x) = \log\left(\frac{-1 + \sqrt{1+4x}}{2}\right)$$

$$\to t = \frac{-1 - \sqrt{1+4y}}{2} \to e^x = \frac{-1 - \sqrt{1+4y}}{2} \quad \text{non acc.}$$

!! ESERCIZIO NON SVOLTO A LEZIONE

2.3) $f(x) = \begin{cases} 2x+1 & , \ x \leq 0 \\ \sqrt{x+1} & , \ x > 0 \end{cases}$

$y = 2x+1 \rightarrow 2x = y-1$

$x = \frac{y-1}{2}$ $\quad x \leq 0 \rightarrow y-1 \leq 0$

$y \leq 1$

$y = \sqrt{x+1}$ $\quad \begin{cases} x \geq 0 \rightarrow y^2 - 1 \geq 0 \\ y \geq 0 \end{cases} \quad \begin{cases} y \leq -1 \cup y \geq +1 \\ y \geq 0 \end{cases}$

$\rightarrow y \geq 1$

$y^2 = x+1$

$x = y^2 - 1$

$$f^{-1}(x) = \begin{cases} \frac{x-1}{2} & , \ x \leq 1 \\ x^2 - 1 & , \ x > 1 \end{cases}$$

3. NUMERI COMPLESSI

3.1 FORMA ALGEBRICA, TRIGONOMETRICA, ESPONENZIALE e INVERSI

$z = a + ib \qquad a, b \in \mathbb{R}$

$= \rho(\cos\theta + i\sin\theta)$

$= \rho e^{i\theta}$

$\rho = \sqrt{a^2+b^2}$

$Re(z) = a$

$Im(z) = b$

a) $\dfrac{1+2i}{1-2i} = \dfrac{1+2i}{1-2i} \cdot \dfrac{1+2i}{1+2i}$

$= \dfrac{1+4i^2+4i}{1-4i^2} = \dfrac{-3+4i}{5} = -\dfrac{3}{5} + \dfrac{4}{5}i$

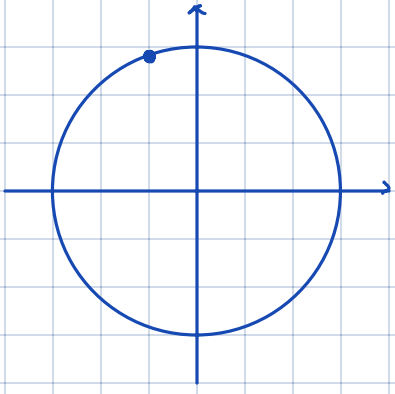
$Re(z) = -\dfrac{3}{5} \qquad Im(z) = +\dfrac{4}{5}$

$|z| = \sqrt{\left(-\dfrac{3}{5}\right)^2 + \left(\dfrac{4}{5}\right)^2} = \sqrt{1} = 1$

$\begin{cases} \cos\theta = -\dfrac{3}{5} \\ \sin\theta = \dfrac{4}{5} \end{cases} \rightarrow \theta = \arccos\left(-\dfrac{3}{5}\right)$

$z = \cos\theta + i\sin\theta$

$= e^{i\theta}$



b) $z = 5e^{-i\frac{\pi}{3}}$ F. ALGEBRICA di z^{-1}

MODO 1 (GIUSTO, ma non consigliato in questo caso)

$z = 5\left(\cos\left(-\dfrac{\pi}{3}\right) + i\sin\left(-\dfrac{\pi}{3}\right)\right) = 5\left(\dfrac{1}{2} - i\dfrac{\sqrt{3}}{2}\right) = \dfrac{5 - i5\sqrt{3}}{2}$

$z^{-1} = \dfrac{2}{5 - i5\sqrt{3}} \cdot \dfrac{5 + i5\sqrt{3}}{5 + i5\sqrt{3}} = \dfrac{10(1 + i\sqrt{3})}{25 + 75} = \dfrac{10(1+i\sqrt{3})}{100_{10}} = \dfrac{1}{10} + \dfrac{\sqrt{3}}{10}i$

MODO 2.

$z = 5e^{-i\frac{\pi}{3}} \qquad z^{-1} = \left(5e^{-i\pi/3}\right)^{-1} = \dfrac{1}{5}e^{i\frac{\pi}{3}} = \dfrac{1}{5}\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)$

$= \dfrac{1}{10} + \dfrac{\sqrt{3}}{10}i$

c) SCRIVI IN F. ALG., TRIG., ESPONENZIALE

$$z = \operatorname{Im}\left(\frac{1}{2\pi i}\cdot\frac{1}{i-2}\right) e^{\frac{2}{3}\pi i}$$

$$= \operatorname{Im}\left(\frac{1}{-2\pi-4\pi i}\right) e^{\frac{2}{3}\pi i}$$

$$= \operatorname{Im}\left(-\frac{1}{2\pi}\;\frac{1}{1+2i}\cdot\frac{1-2i}{1-2i}\right) e^{\frac{2}{3}\pi i}$$

$$= \operatorname{Im}\left(-\frac{1}{2\pi}\;\frac{1-2i}{5}\right) e^{\frac{2}{3}\pi i}$$

$$= \operatorname{Im}\left(-\frac{1}{10\pi}+\frac{1}{5\pi}i\right)\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)$$

$$= \frac{1}{5\pi}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) \longrightarrow -\frac{1}{10\pi}+\frac{\sqrt{3}}{10\pi}i = z \quad \text{F. ALG.}$$

$$\frac{1}{5\pi}\left(\cos\frac{2}{3}\pi + i\sin\frac{2}{3}\pi\right) = z \quad \text{F. TRIG.}$$

$$\frac{1}{5\pi}\, e^{i\frac{2}{3}\pi} = z \quad \text{F. ESP.}$$

d) Determina $z \in \mathbb{C}$ t.c.

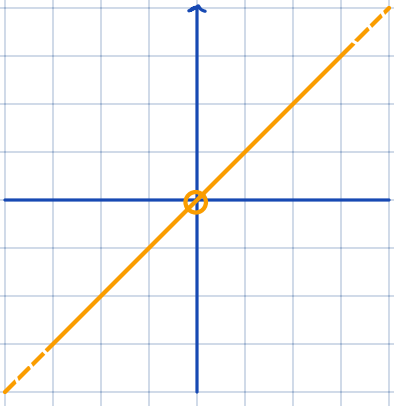
$$\frac{\operatorname{Re}(z)}{z} = \frac{1}{2} - \frac{1}{2}i \qquad z \neq 0$$

$$2\operatorname{Re}(z) = (1-i)z \qquad z = x+iy \quad x, y \in \mathbb{R}$$

$$2\operatorname{Re}(x+iy) = (1-i)(x+iy)$$

$$2x = x + iy - ix + y$$

$$(x-y) + i(x-y) = 0 \rightarrow \begin{cases} x-y=0 \\ x-y=0 \end{cases} \quad x \neq 0, y \neq 0 \qquad \begin{cases} x=y \\ x=y \end{cases} \quad x \neq 0, y \neq 0$$



3.2 POTENZE

$z^n = (a+ib)^n$ POTENZA di un BINOMIO

$= [\rho(\cos\theta + i\sin\theta)]^n = \rho^n(\cos(n\theta) + i\sin(n\theta))$ DE MOIVRE

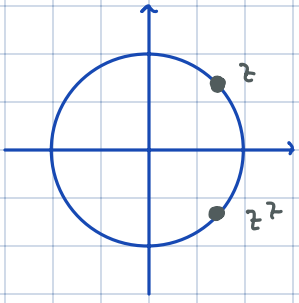
$= (\rho e^{i\theta})^n = \rho^n e^{i(n\theta)}$

a) $z = \sqrt{2} \cdot \frac{1+i}{2} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$ $\quad \rho = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$

$z^7 = ?$ $\quad = 1 \cdot \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i\right)$ $\quad \begin{cases} \frac{\sqrt{2}}{2} = \cos\theta \\ \frac{\sqrt{2}}{2} = \sin\theta \end{cases} \rightarrow \theta = \frac{\pi}{4}$

$= \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$

$z^7 = \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^7 = \cos\frac{7}{4}\pi + i\sin\frac{7}{4}\pi = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$



b) ESERCIZIO 3 FOGLIO

$(1+i)^5$

$z = 1+i$

$|z| = \sqrt{1^2+1^2} = \sqrt{2}$

$z = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i\right) = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

$z^5 = \sqrt{2^5}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$

$= 4\sqrt{2}\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = -4 - 4i$

3.3 RADICI

$z^n = w \qquad w, z \in \mathbb{C}$

$w = \rho_w e^{i\theta_w} \qquad z = \rho_z e^{i\theta_z}$

$z^n = \rho_w e^{i\theta_w} \rightarrow \rho_z^n e^{i\theta_z n} = \rho_w e^{i\theta_w}$

$$\begin{cases} \rho_z^n = \rho_w \rightarrow \rho_z = \sqrt[n]{\rho_w} \\ \theta_z \cdot n = \theta_w + 2k\pi \\ \quad \hookrightarrow \theta_z = \dfrac{\theta_w + 2k\pi}{n} \end{cases}$$

$$\boxed{z = \sqrt[n]{\rho_w}\, e^{i\frac{\theta_w + 2k\pi}{n}}} \qquad k = 0, \dots, n-1$$

a) RADICI CUBICHE di $w = 2(i-1) = -2 + 2i = 2\sqrt{2}\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)$

$\rho_w = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$

$= 2\sqrt{2}\, e^{i\frac{3}{4}\pi}$

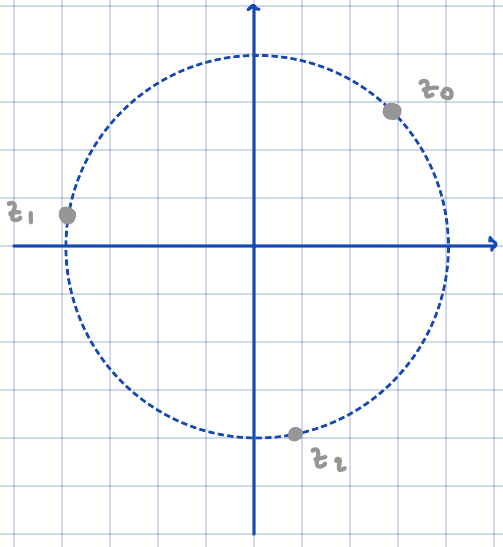
$= \sqrt{8}\, e^{i\frac{3}{4}\pi}$

$z = \sqrt[6]{8}\, e^{i\frac{\frac{3}{4}\pi + 2k\pi}{3}} \qquad k = 0, 1, 2$

$k=0: \quad z_0 = \sqrt[6]{8}\, e^{i\frac{\pi}{4}} = \sqrt[6]{8}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)$

$k=1: \quad z_1 = \sqrt[6]{8}\, e^{i\frac{11}{12}\pi} = \sqrt[6]{8}\left(\cos\frac{11}{12}\pi + i\sin\frac{11}{12}\pi\right)$

$k=2: \quad z_2 = \sqrt[6]{8}\, e^{i\frac{19}{12}\pi} = \sqrt[6]{8}\left(\cos\frac{19}{12}\pi + i\sin\frac{19}{12}\pi\right)$



$(r = \sqrt[6]{8})$

b) CALCOLA LE RADICI QUARTE di $\frac{1}{5\pi}\left(\cos\left(\frac{2}{3}\pi\right)+i\sin\left(\frac{2}{3}\pi\right)\right)$

$$w=\frac{1}{5\pi}\left(\cos\left(\frac{2}{3}\pi\right)+i\sin\left(\frac{2}{3}\pi\right)\right) \qquad \text{oss: } |w|=\frac{1}{5\pi}$$

$$z^4=w \Rightarrow z=\sqrt[4]{\frac{1}{5\pi}}\left(\cos\frac{\frac{2}{3}\pi+2k\pi}{4}+i\sin\frac{\frac{2}{3}\pi+2k\pi}{4}\right) \qquad k=0,1,2,3$$

$$z_0=\sqrt[4]{\frac{1}{5\pi}}\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)=\sqrt[4]{\frac{1}{5\pi}}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right)$$

$$z_1=\sqrt[4]{\frac{1}{5\pi}}\left(\cos\frac{2}{3}\pi+i\sin\frac{2}{3}\pi\right)=\sqrt[4]{\frac{1}{5\pi}}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)$$

$$z_2=\sqrt[4]{\frac{1}{5\pi}}\left(\cos\frac{7}{6}\pi+i\sin\frac{7}{6}\pi\right)=\sqrt[4]{\frac{1}{5\pi}}\left(-\frac{\sqrt{3}}{2}-\frac{1}{2}i\right)$$

$$z_3=\sqrt[4]{\frac{1}{5\pi}}\left(\cos\frac{5}{3}\pi+i\sin\frac{5}{3}\pi\right)=\sqrt[4]{\frac{1}{5\pi}}\left(\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)$$

!! ESERCIZIO NON SVOLTO A LEZIONE

3.4) EQUAZIONI

a) $2z^2 + \bar{z} = -1$

$z = x + iy \qquad x, y \in \mathbb{R}$

$2(x+iy)^2 + (x-iy) = -1$

$2(x^2 - y^2 + 2ixy) + x - iy = -1$

$2x^2 - 2y^2 + 4ixy + x - iy + 1 = 0$

$(2x^2 - 2y^2 + x + 1) + i(4xy - y) = 0 \rightarrow \begin{cases} 2x^2 - 2y^2 + x + 1 = 0 \\ 4xy - y = 0 \end{cases}$

$\begin{cases} y(4x-1) = 0 \end{cases}$

$y = 0 \rightarrow \begin{cases} 2x^2 + x + 1 = 0 \\ y = 0 \end{cases} \quad \begin{cases} x = \frac{-1 \pm \sqrt{-7}}{4} \notin \mathbb{R} \end{cases} \rightarrow \phi$

$x = \frac{1}{4} \rightarrow \begin{cases} \frac{1}{8} - 2y^2 + \frac{5}{4} = 0 \\ x = \frac{1}{4} \end{cases} \quad \begin{cases} 2y^2 = \frac{11}{8} \\ x = \frac{1}{4} \end{cases} \quad \begin{cases} y = \pm \frac{\sqrt{11}}{4} \\ x = \frac{1}{4} \end{cases}$

$z_1 = \frac{1}{4} + i\frac{\sqrt{11}}{4} \qquad z_2 = \frac{1}{4} - i\frac{\sqrt{11}}{4}$

b) $z^5 = -16\bar{z}$ $\qquad z = (x+iy)$ $\qquad z^5 = (x+iy)^5$

TROVIAMO ρ

$|z^5| = |-16\bar{z}|$

$|z|^5 = 16|z|$

$\rho^5 = +16\rho$

$|z^5| = |z|^5 = \rho^5$

$|-16\bar{z}| = |-16| \cdot |\bar{z}| = 16|\bar{z}| = 16|z| = 16\rho$

$\rho^5 - 16\rho = 0 \rightarrow \rho(\rho^4 - 16) = 0$

$\rho = 0 \rightarrow z = 0$

$\rho^4 = 16 \rightarrow \rho = -2$ non acc.

$\rho = 2$

$\rightarrow z = 2e^{i\theta}$

$(2e^{i\theta})^5 = -16 \cdot 2e^{-i\theta}$

$32e^{i5\theta} = -32e^{-i\theta}$

$e^{i5\theta} = -e^{-i\theta}$

$e^{i6\theta} = -1$ [RADICI SESTE DELL'UNITÀ]

$e^{i6\theta} = e^{i(\pi + 2k\pi)}$

$6\theta = \pi + 2k\pi \rightarrow \theta = \frac{\pi + 2k\pi}{6}$

$\rightarrow z = 2e^{i\frac{\pi+2k\pi}{6}} \qquad k = 0, \dots, 5$

$z = 0 \quad \cup \quad z = 2\, e^{i\frac{\pi + 2k\pi}{6}} \qquad (k = 0, \dots, 5)$

MODO 2: $z^5 = -16\bar{z}$ $\qquad \rho = 0 \Rightarrow z = 0$ è SOLUZIONE

$\rho = 2$

$z^5 \cdot z = -16\, z \cdot \bar{z}$

$z^6 = -16 |z|^2$

$z^6 = -16 \cdot 4$

$z^6 = -64$

$z^6 = 2^6 e^{i(\pi + 2k\pi)}$

$z\bar{z} = |z|^2$

$\uparrow$

$(x - iy)(x + iy)$

$=$

$x^2 + y^2$

c) $\dfrac{z^4}{z^4 + 1} = 1 - \dfrac{i}{\sqrt{3}}$

!! ESERCIZIO NON SVOLTO A LEZIONE

$\sqrt{3}\, z^4 = (\sqrt{3} - i)(z^4 + 1)$

$\sqrt{3}\, z^4 = \sqrt{3}\, z^4 + \sqrt{3} - i z^4 - i$

$\dfrac{i\, z^4}{i} = \dfrac{\sqrt{3} - i}{i}$

$z^4 = \dfrac{\sqrt{3}}{i} - 1$

$z^4 = -\sqrt{3}\, i - 1$

$w = -1 - \sqrt{3}\, i$

$= 2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i\right)$

$= 2\, e^{i\frac{4}{3}\pi}$

$|w| = \sqrt{1 + 3} = 2$

$z_k = \sqrt[4]{2}\; e^{i\frac{\frac{4}{3}\pi + 2k\pi}{4}}, \quad k = 0, \dots, 3$

$k = 0: \; z_0 = \sqrt[4]{2}\, e^{i\frac{\pi}{3}} = \sqrt[4]{2}\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$

$k = 1: \; z_1 = \sqrt[4]{2}\, e^{i\frac{5}{6}\pi} = \sqrt[4]{2}\left(-\frac{\sqrt{3}}{2} + i\frac{1}{2}\right)$

$k = 2: \; z_2 = \sqrt[4]{2}\, e^{i\frac{4}{3}\pi} = \sqrt[4]{2}\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$

$k = 3: \; z_3 = \sqrt[4]{2}\, e^{i\frac{11}{6}\pi} = \sqrt[4]{2}\left(\frac{\sqrt{3}}{2} - \frac{1}{2} i\right)$

